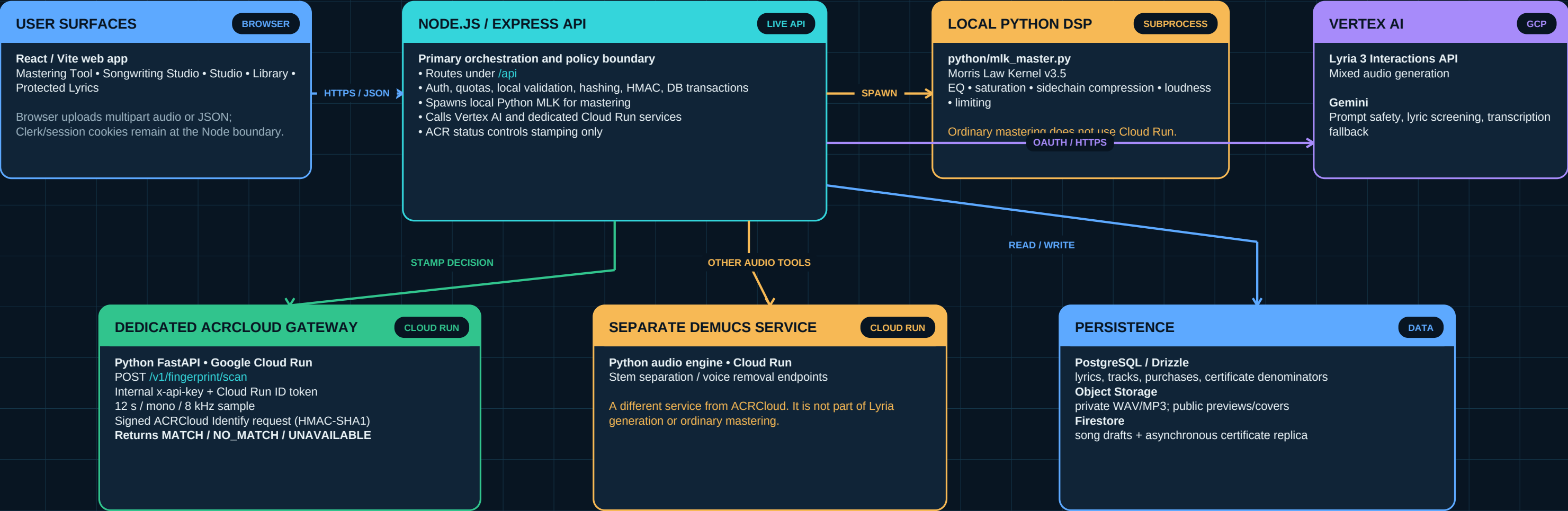


GravelKing Pro — Live Backend Pipeline Architecture

Exact service boundaries, implemented routes, persistence layers, and failure isolation



NON-NEGOTIABLE AVAILABILITY INVARIANT

IMPLEMENTED

Every tool completes independently. ACRCLOUD can suppress a stamp; it cannot suppress the audio result.

NO_MATCH → eligible path creates the dual-anchor certificate and embeds the LSB nominator.

MATCH → processing / generation / storage continues; no certificate row, no LSB payload.

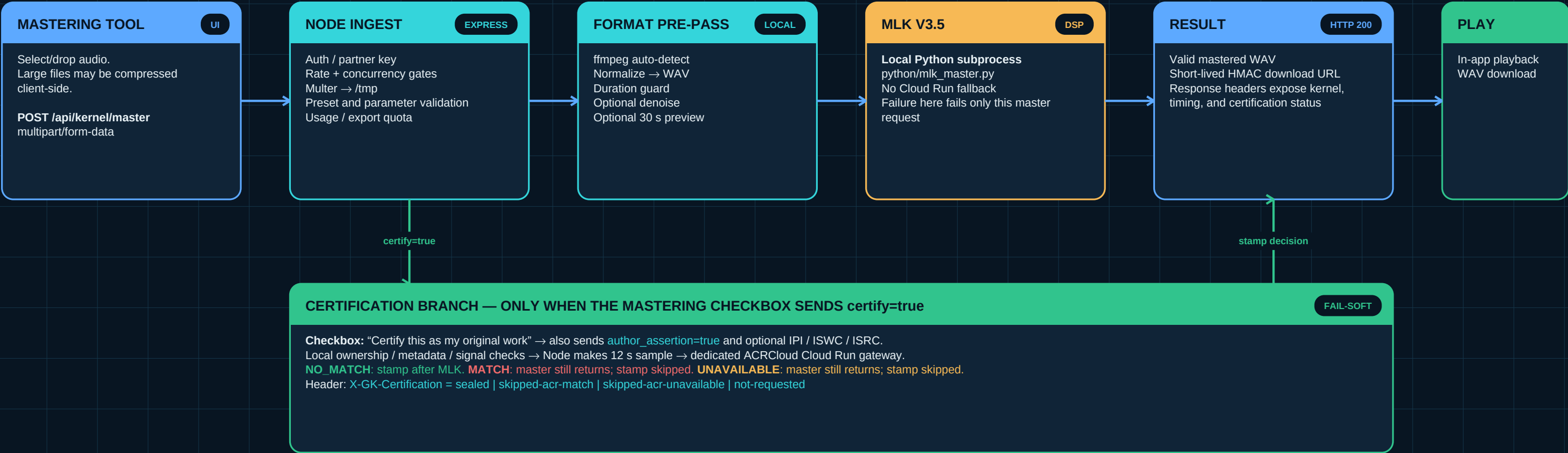
UNAVAILABLE (timeout, 401, 402, 429, 5xx, billing suspension, missing config) → processing / generation / storage continues; no stamp.

Ordinary mastering bypasses ACRCLOUD completely unless the user explicitly requests certification.

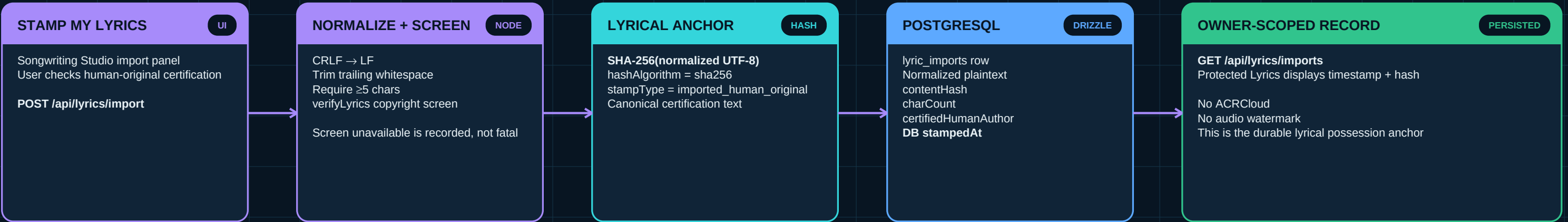
Scenario Pipelines — Upload/Mastering + Original Lyrics

Exact request paths and branch behavior

1 Audio Upload & Mastering



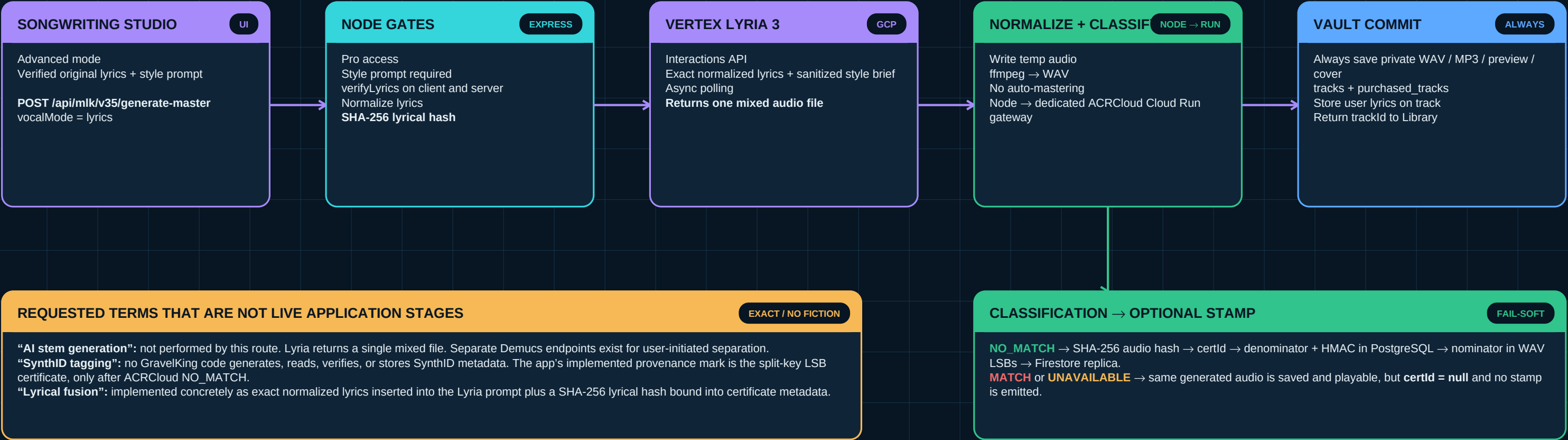
2 Original Lyrics First



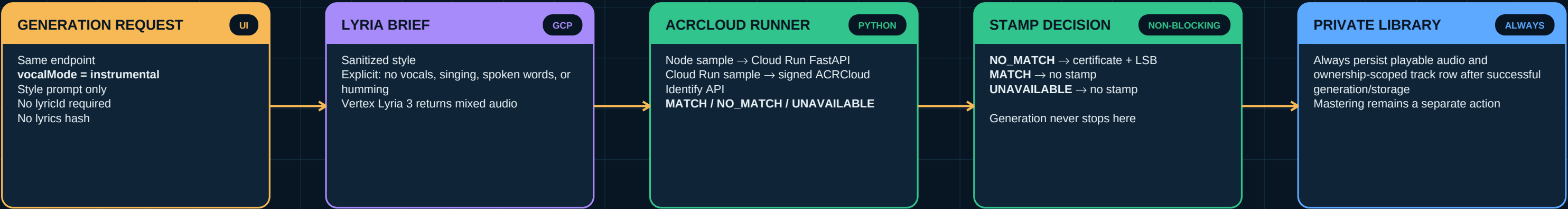
Scenario Pipelines — AI Song + Instrumental Generation

Lyria produces mixed audio; ACRCLOUD is non-blocking and controls only the stamp

3 Lyrics + AI Instrumental / Song Generation



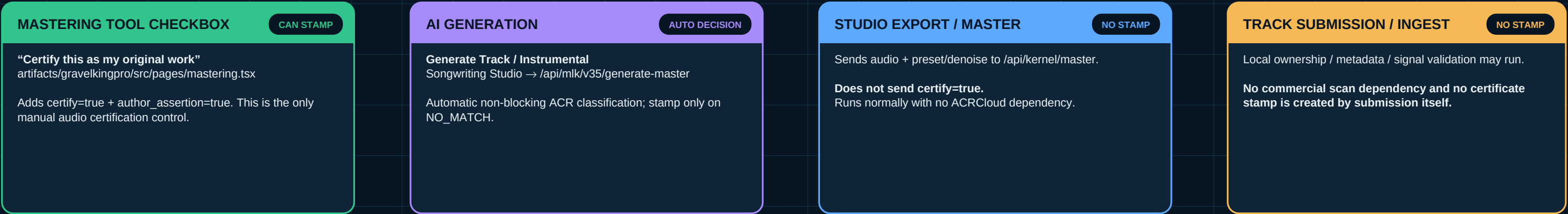
4 Pure Instrumental AI Generation



Certificate Stamping Engine — Triggers, Gates, and Isolation

What can request a stamp, what cannot, and exactly how every ACRCLOUD state resolves

5 Actual UI Triggers



A Stamp Gate State Machine

